# Econ 2 - Lecture 9 - 4/29/26

## Midterm Exam

Median = 25/30 Higher than usual!

Final Exam material is more difficult → End-of-Year fatigue

Discussion Section this week = Exam Review

No Lecture Quiz this week

Note: Friday Review Session this week (5/1), not next week (5/8)

Weeks 6-10: Lecture Quizzes + 3 more activities

Last Wednesday: 2nd - Half of Quarter Goal

Find Macroeconomic Equilibrium ⇒ $Y = AE$

$Y$ = GDP (Chapter 3.1) = $C + I + G + NX$

AE = Aggregate Expenditures = Spending

Biggest Component of AE = Consumption ($C$)

Biggest Determinant of $C$ = Income ($Y$)

## Consumption Drivers

1) Income ⇒ Paycheck → Taxes
($Y$) ($T$)

$\underbrace{\text{Disposable Income}}_{=\text{Take-Home Income}}$ (DI) = $Y - T$

If DI ↑ ↗ $Y\uparrow$ ↘ / ↘ $T\downarrow$ ↗ $C\uparrow$

If DI ↓ ↗ $Y\downarrow$ ↘ / ↘ $T\uparrow$ ↗ $C\downarrow$

2.) Wealth = Value of what we own − Value of what we owe

Assets − Liabilities

- → Stocks, Bonds
- → Home
- → Car
- → Land
- → Personal IP
- → Fine Art
- → ~~NFTS?~~ Law
- → Gold

- → Mortgages
- → Student Loans
- → Car Loans
- → Credit Card

As wealth ↑, C ↑, wealth ↓, C ↓

3.) Interest Rate (r)

Cost of borrowing, benefit of saving

If r increases to 20% → C ↓

If r ↓ ⇒ C ↑

4.) Expectations: What is next year looking like?

If next year looks poor → geopolitical issues

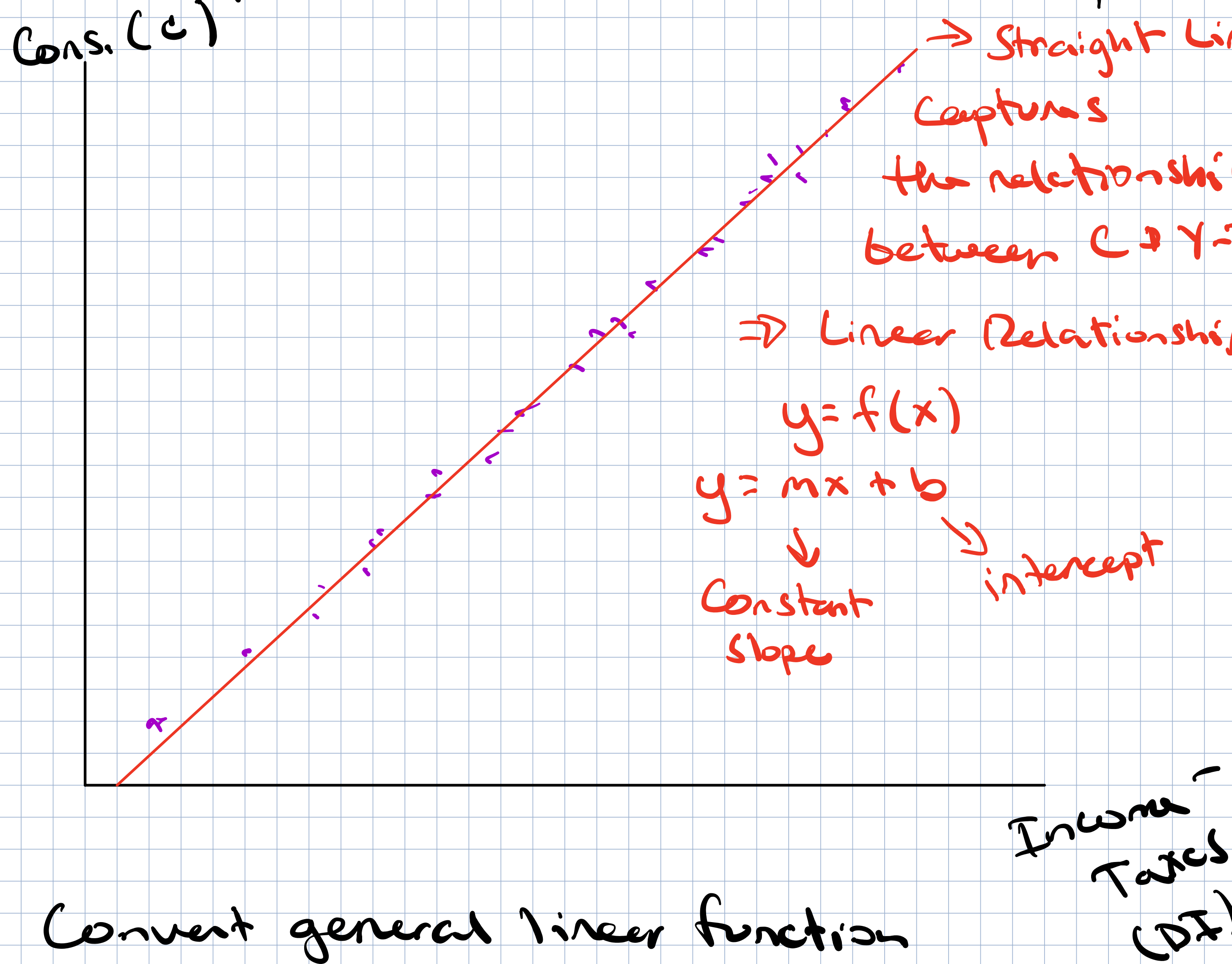
↓

C ↓ today

5.) Preferences: Kids, family size

→ Pandemic Lockdown

How important is income (Y) to Consumption (C)?




→ Straight Line captures the relationship between C & Y-T

⇒ Linear Relationship

$y = f(x)$

$y = mx + b$

$m$ → Constant slope

$b$ → intercept

Convert general linear function into consumption function

$$C = \text{Intercept} + \text{Slope} \times (Y - T)$$

What does the intercept represent?

Value of outcome when $Y - T = \emptyset$?

Level of C when $Y - T = \emptyset$

↳ food, water, utilities, healthcare, necessities

Baseline level of spending regardless of $Y - T$

⇒ Win lottery, income is unchanged

⇒ Wealth ↑

⇒ Spending ↑, C ↑

Larger intercept!

Intercept = Autonomous Consumption (AC)

↳ Source of funding if $Y - T = 0$

⇒ Savings, borrow, Gov't subsidy, family, etc.

What about slope of our consumption function?

$$\text{Slope} = \frac{\text{Rise}}{\text{Run}} = \frac{\text{Change in } C}{\text{Change in } Y-T} = \frac{\Delta C}{\Delta (Y-T)}$$

Constant Slope = Linear function → straight line

Same increase in C for every additional take-home dollar increase (DI ↑ by \$1)

Slope ⇒ Fraction of Y-T we put towards C

Consumption function

$$C = \text{Autonomous Consumption} + \text{Marginal Propensity to Consume} \cdot (Y - T)$$

$$C = AC + \underbrace{MPC\,(Y - T)}_{\text{Fraction of Take-Home Income we Spend}}$$

$AC$ ↓ "Automatic" C

Fraction of Take-Home Income we Spend

↙ Spend MPC

↘ Save MPS (save)

1 − MPC

Table of Consumption

| $Y-T$ | $C$ |
| --- | --- |
| 0 | 2000 |
| 2000 | 3200 |
| 4000 | 4400 |
| 6000 | 5600 |
| 8000 | 6800 |
| 10000 | 8000 |

AC = \$2000

C = 2000 when $Y-T = \emptyset$

What is mpc?

Increase Y-T by \$2000

Increase C by \$1200

If Y-T increases by \$1000, C increase by \$600

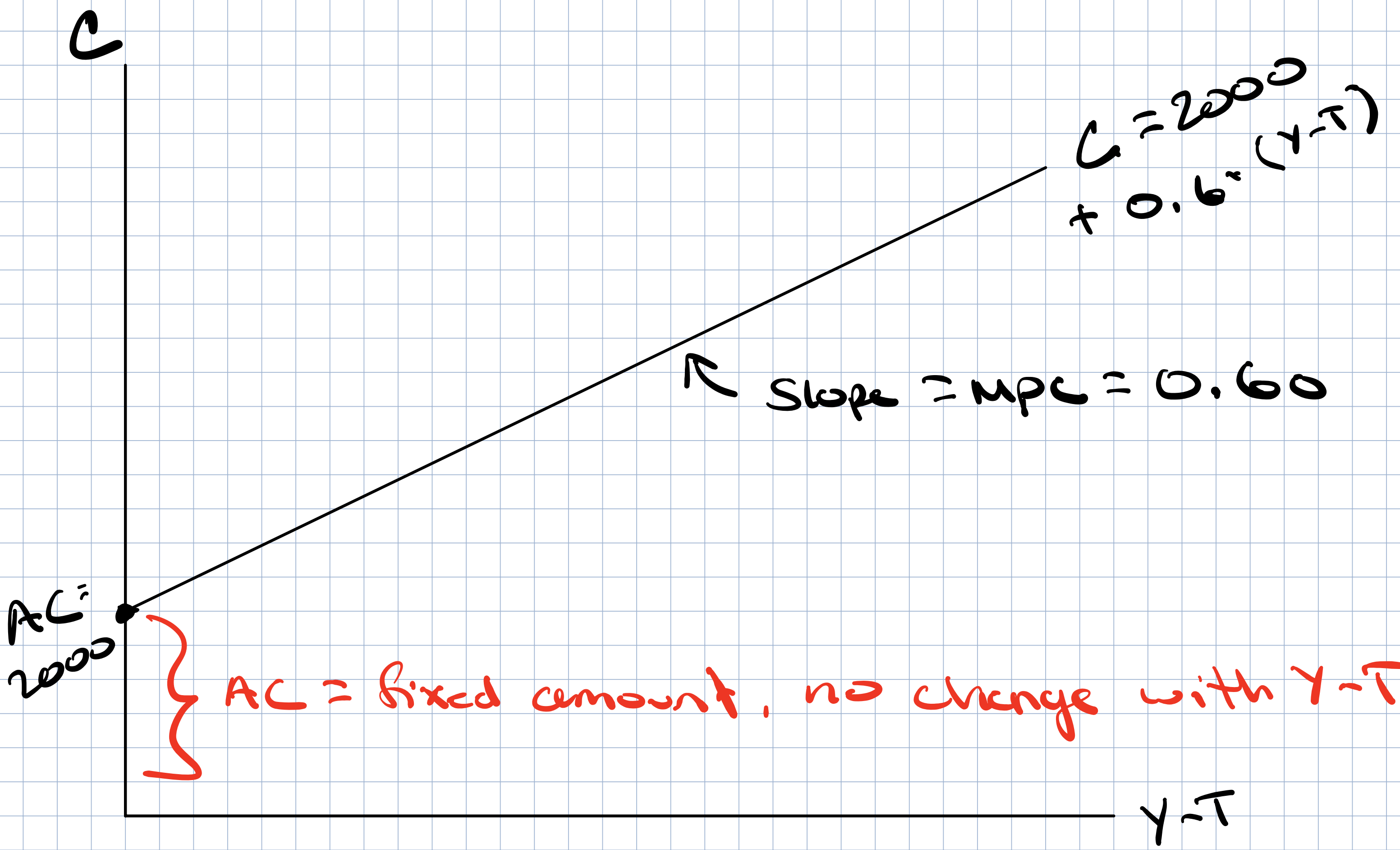
If Y-T increases by \$100, C increase by \$60

Increase Y-T by \$1, increase C by \$0.60

$$mpc = \frac{\Delta C}{\Delta Y-T} = \frac{1200}{2000} = \frac{0.6}{1} = 0.60$$

$$C = 2000 + 0.6 \times (Y - T)$$

Graph $C = 2000 + 0.6 * (Y - T)$




Goal is finding where $Y = AE$

Not looking for $Y - \cancel{T} = AE$

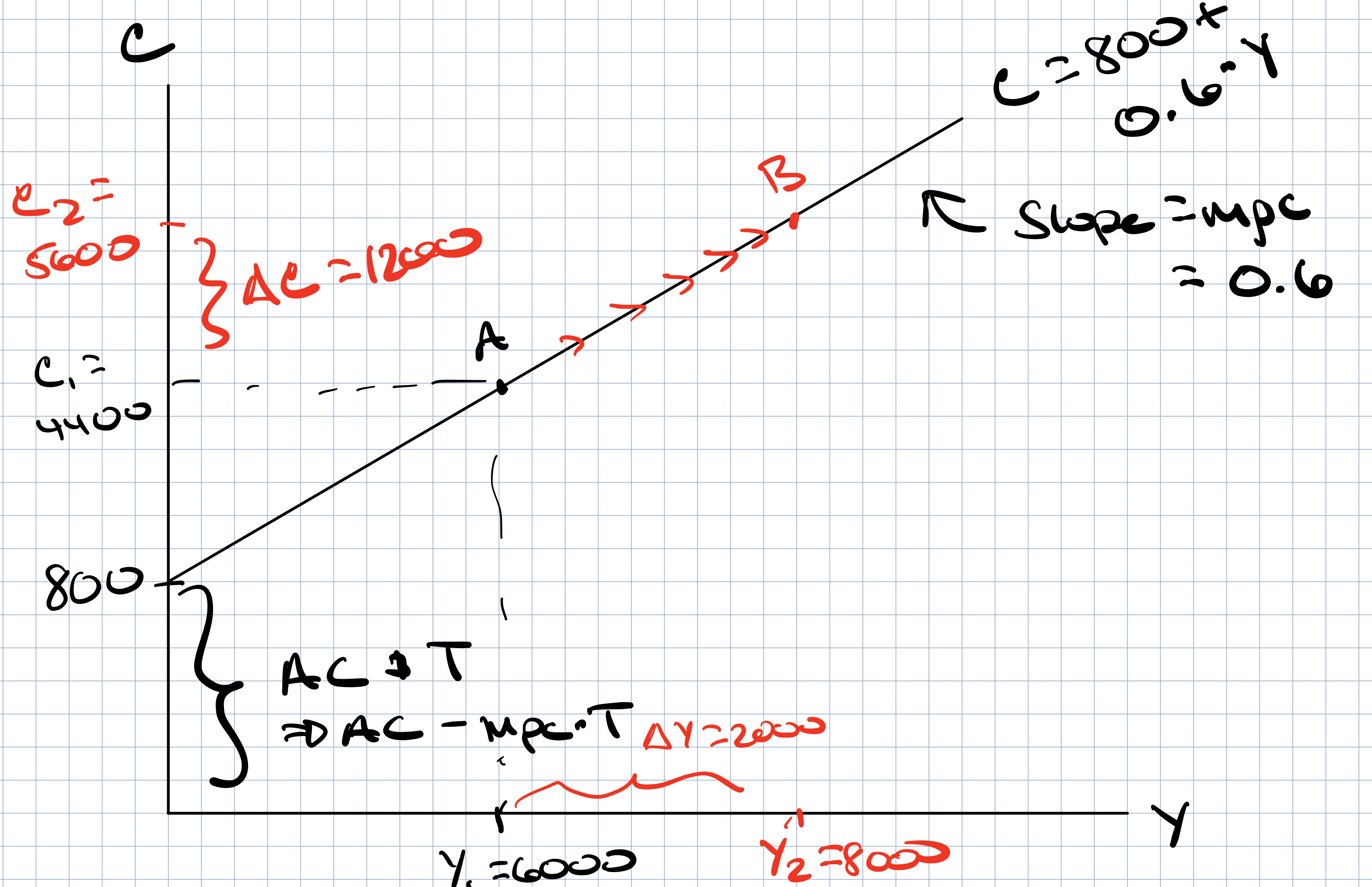
Account for taxes (T) ⇒ Assume T is fixed

↳ Same value of T for all levels of Y → Set T = 2000

| Y | T | Y - T | C |
|---|---|---|---|
| 2000 | 2000 | 0 | 2000 |
| 4000 | 2000 | 2000 | 3200 |
| 6000 | 2000 | 4000 | 4400 |
| 8000 | 2000 | 6000 | 5600 |
| 10000 | 2000 | 8000 | 6800 |
| 12000 | 2000 | 10000 | 8000 |

Convert to equation:

$$C = AC + mpc \cdot (Y - T)$$
$$C = 2000 + 0.6\,(Y - 2000)$$
$$C = 2000 + 0.6 \cdot Y - (0.6 \times 2000)$$
$$C = 800 + 0.6 \cdot Y$$




If $Y_1 = 6000 \rightarrow Y_2 = 8000$

Move along the consumption function.

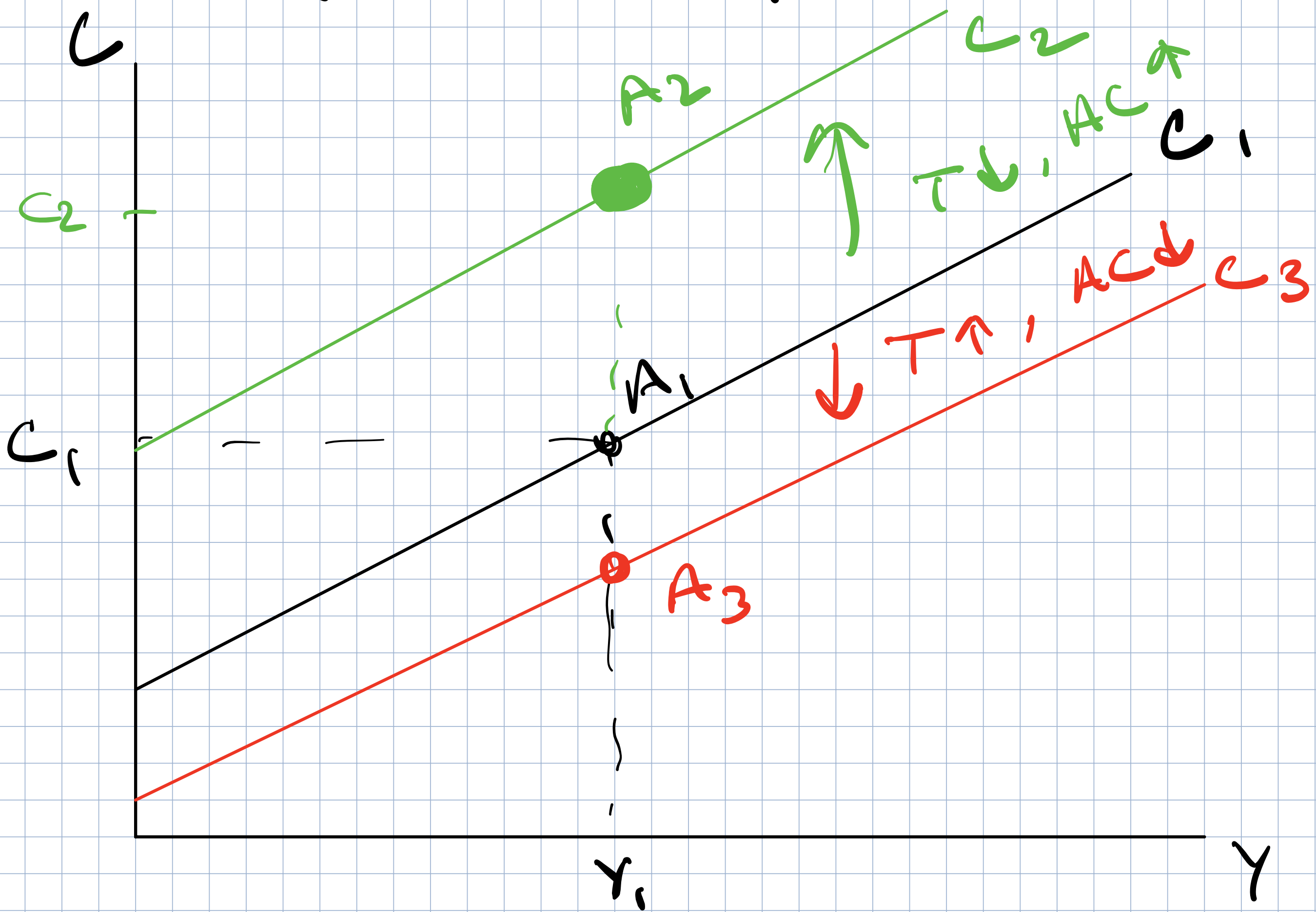
Do not shift C−function when Y changes

What will change the entire consumption function?

What will change consumption other than income?

What shifts consumption?

Shifter of $C = AC + mpc \cdot (Y - T)$




Two Big Shifters of C

1. Taxes (T)

If T↑ ⇒ Y−T↓ = C↓ }
If T↓ ⇒ Y−T↑ = C↑ } No change in Y

2. Autonomous Consumption (AC)

More kids → C↑ at all Y

Stock Market Crash ⇒ C↓ at all Y

⇒ Changes in AC